

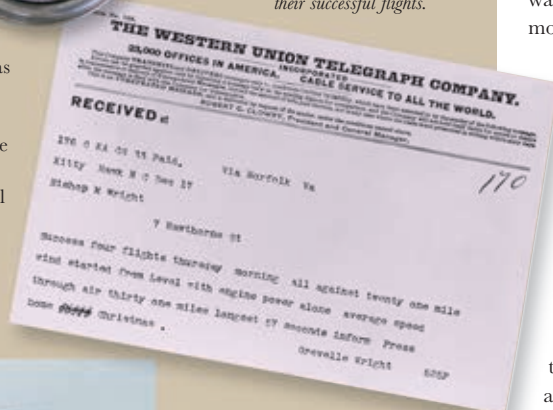
and was traveling on a fairly even course.” Wilbur flew for 59 seconds, traveling a distance of 852ft (260m), before the Flyer pitched down to a bone-jarring landing, breaking the elevator support. The Flyer has spent less than a minute in the air, but it was enough to constitute sustained, controlled, powered flight!

Shortly after this momentous event, the Flyer was caught by a gust of wind as it was being carried back to the camp and rolled over, taking with it Daniels, who was lucky to end the day with no more than cuts and bruises. The flying machine was a wreck, but the brothers did not let the incident spoil their delight in their achievement. After lunch the brothers walked over to the Kitty Hawk weather station to telegraph home the news of their success and imminent return.



CAUGHT IN TIME

This handheld stopwatch was used to time the four flights made by the Wrights on December 17, 1903. Below is the understated telegram that Orville sent their father asking him to inform the press of their successful flights.



HARD LANDING

At the end of the fourth flight, which lasted for almost a minute, Wilbur landed hard and broke the Flyer's elevator support. It was then transported back to the Wrights' campsite, where they intended to repair the damage.



The new data led the Wrights to design a wing that was longer and slimmer, with a flatter camber. For the first time they added a tail – two fixed vertical fins that they hoped would prevent the machine from going out of control in banked turns.

The 1902 No. 3 glider proved that the Wrights' calculations were correct. Aerodynamically, it was the most efficient machine yet built, but at first it was even more tricky to control than their previous model. After a few dangerous spills, the Wrights

took stock and came up with a solution. The

culprit was the fixed tailfins. They needed to be movable so they could be turned to counterbalance drag. With typical ingenuity, the brothers created a control system that linked the rudder to the wing-warping mechanism. By the summer's end they were making controlled glides of up to 600ft (200m), staying airborne for up to 26 seconds.

Now the Wrights were ready to embark on the momentous step to powered flight. For this they needed an engine and a propeller. When automobile companies proved incapable of supplying a suitable engine, the Wrights had one made by their assistant Charlie Taylor, who delivered a remarkable gasoline engine weighing 180lb (82kg) and delivering 12hp. In contrast, the problem of propeller design proved astonishingly complex, forcing the brothers to tackle intricate questions of theoretical physics and mathematics.

The Wrights returned to Kill Devil Hills in late September 1903, well aware that, at that very moment, Samuel Pierpont Langley was preparing for the first flight of his Great Aerodrome. When news came through that his first attempt had failed, Wilbur wrote: "It seems to be our turn to throw now, and I wonder what our luck will be."

For a time, luck seemed to be against them. In stationary tests, the engine proved temperamental and eventually damaged the propeller shafts. They were sent for repair, but when tests resumed at the end of November, one of the repaired shafts was found to be cracked. Orville returned to Dayton to make completely new steel shafts.



PRESS MISREPRESENTATION

Typical of the stories circulating in the days following the Kitty Hawk flights was this headline from the Norfolk Virginian-Pilot. Such wild inaccuracies provoked the brothers into issuing a press release containing exact details of their achievement.